DO YOU KNOW?

The principle behind the working of a parabolic mirror has been known since the Greco-Roman times. The first mention of these structures was found in the book, 'On Burning Mirrors', written by the mathematician Diocles. They were also studied in the 10th century, by a physicist called Ibn Sahl. The first parabolic mirrors were constructed by Heinrich Hertz, a German physicist, in the form of reflector antennae in the year 1888.

3.2 Terms related to Spherical Mirrors

In order to understand the image formation in spherical mirrors, we need to know about some of the terms related to them.

Center of Curvature

It is the center of the sphere from which the mirror is made. It is denoted by the letter **C** in the ray diagrams. (A ray diagram represents the formation of an image by the spherical mirror. You will study about them in the higher classes).

Pole

It is the geometric centre of the spherical mirror. It is denoted by the letter **P**.

Radius of Curvature

It is the distance between the center of the sphere and the vertex. It is shown by the letter **R** in ray diagrams (*The **vertex** is the point on the mirror's surface where the principal axis meets the mirror.* It is also called as '**pole**').

Principal Axis

The line joining the pole of the mirror and its center of curvature is called principal axis.

Focus

When a beam of light is incident on a spherical mirror, the reflected rays converge (concave mirror) at or appear to diverge from (convex mirror) a point on the principal axis. This point is called the 'focus' or 'principal focus'. It is also known as the focal point. It is denoted by the letter **F** in ray diagrams.

Focal length

The distance between the pole and the principal focus is called focal length (f) of a spherical mirror.

There is a relation between the focal length of a spherical mirror and its radius of curvature. The focal length is half of the radius of curvature.

$$\text{Focal length} = \frac{\text{Radius of curvature}}{2}.$$

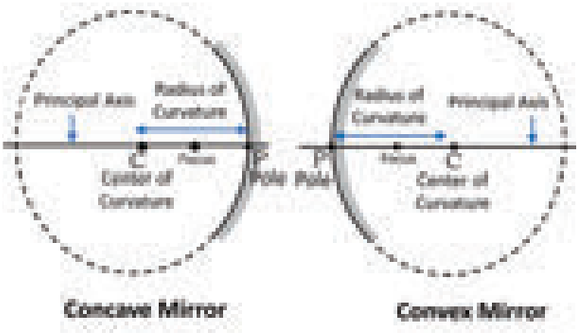


Figure 3.3 Terms related to a spherical mirror

Problem 1

The radius of curvature of a spherical mirror is 20 cm. Find its focal length.

Solution

Radius of curvature = 20 cm

$$\text{Focal length (f)} = \frac{\text{Radius of curvature}}{2}$$

$$= \frac{R}{2} = \frac{20}{2} = 10 \text{ cm}$$

Problem 2

Focal length of a spherical mirror is 7 cm. What is its radius of curvature?

Solution

Radius of curvature (R) = 2 × Focal length

= 2 × 7 = 14 cm

8th_Science_Unit-3_EM.indd 24 03-03-2020 15:10:01